MKALAMA DISTRICT COUNCIL

FORM THREE TERMINAL DISTRICT EXAMINATION

MARKING GUIDE -BIOLOGY 2025

1.

I	ii	iii	iv	v	vi	vii	viii	ix	x
C	C	A	B	B	C	E	C	B	D

1@10

2.

I	ii	iii	iv	v	vi
G	A	E	C	B	F

SECTION B (54 MARKS)

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3. i. used for grinding or crushing substances

ii. used for keeping specimens under observation

iii. used for magnifying specimens

iv.used for measuring volume of liquids(1.5@=09marks)

iv. used for holding chemicals during experiments

v. used for holding test tubes during heating

4. a) (i) -water

- carbon dioxide

-chlorophyll

-sunlight. **Any 4 point @ ½ = 2 marks**

(ii) – provides nutrient to cells

- getting rid of wastes
- protecting joints and organs
- maintain body temperature
- water keeps our bodies hydrated, allowing cell to function properly.
- Water is the primary component of the blood which carries the oxygen and nutrients though the body

Any 3 points @ 1 mark =3marks

(b) (i) hazardous waste, are the waste which pose immediate threat to human health to the environment

½ mark

Example: medical wastes **½ marks = 01 mark**

Non-hazardous waste, are those waste which does not pose immediate threat to human health and environment **½ marks**

Example: plastic bottles **½ marks = 1 mark**

(ii) – reusing

- Recycling
- Reducing

Any 2 points @ 1 mark= 2 marks

5. (i) Reflex action **01mark**

(ii) Reflex arc **01 mark**

(iii) A well labeled diagram to show reflex aic***03 marks***

- During hot weather, much water is lost from the body through sweating making an increase in blood osmotic pressure, later this causes an increases rate of water reabsorption into the body to replace the lost one.

01mark

- The type of foods taken; Liquid foods increases the amount of water in the body white sugary and salty foods result into decrease in the amount ofwater in the body.

01mark

-Secretion of antiduretic hormone (ADH) by the pituitary gland; The presence of antiduretic hormone in blood increases permeability of the kidney to water which results into reabsorption of water into the body.

01mark

(iv) It helps organism to do away with any sudden stimulus which would cause to the body. **01mark**a)

6 (a)similarities of division coniferophyte and division angiospermophyta

i. both are seeds producing plants

ii.both have true roots, stems and leaves

iii. both have vascular bundles

any 3 three points @ 1.5 marks = 4.5marks

b) Differences between division coniferophyte and angiospermophyta

ANGIOESPERMS	GYMONOSPERMS(CONFERS)
i. they produce fruits that contain seeds	-they produce cones that contain seed
ii. they are flowering plants	-they are non-flowering plants
iii. their seeds are enclosed in an ovary	-their seeds are endo sed in cons
iv. their reproductive structure are flowers	-their reproductive structure are cons

any 3 three points @ 1.5marks = 4.5 marks

07.(a)function of the skeleton

i. support, ie it provide a framework that support the bodies'weight
ii. protection, ie protect heart and lung example ribs
iii. movement, ie bone act as leveler for muscle to pull on allowing movement
iv. blood cell production, example RBCs and WBCs produces from bone marrow
v. mineral storage, ie bones store mineral like calcium and phosphorus

any 5 points @ 1 mark = 5

(b)

Skeletal muscles	**Cardiac muscles**

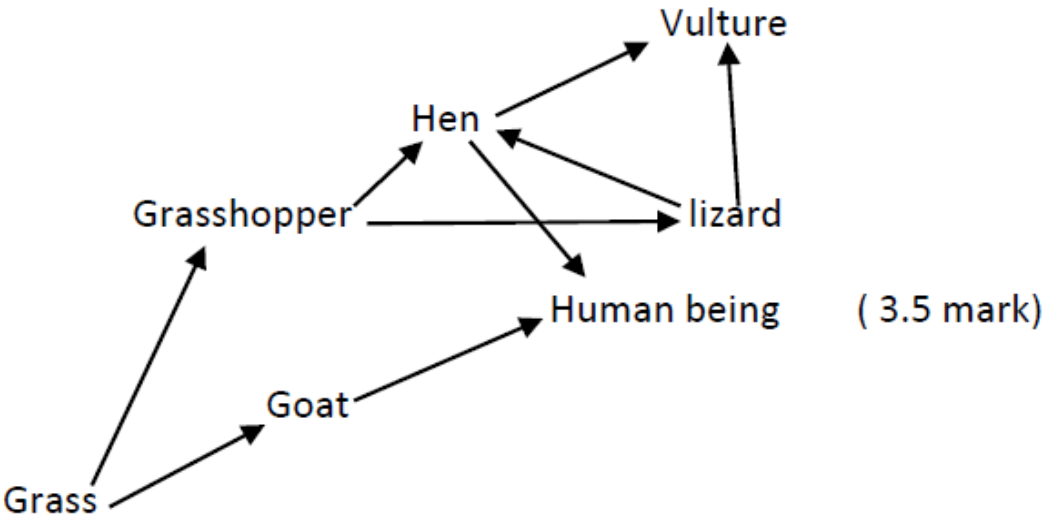
Found on the skeleton	Found in the heart
Get fatigue	Does not get fatigue
Controlled by voluntary nervous system	Controlled by involuntary nervous system

Any 2 point @ 1mark = 2marks

(c) -articular cartilage **1 mark**

- synovial fluid **1 marks**

08.(i) Food web



(3.5 mark)

(ii) (a) producer(s) Grass **(1.5 mark)**
(b) Primary consumers Goat, grasshopper **(1.5 mark)**
(iii) Decomposers (Bacteria or Fungi) **(1.5 mark)**

(b) (i) Parasitism is the type of symbiotic relationship between two organisms the parasite and
host. Mosquito and human being and tick and cow (1 mark)

SECTION C

09.The student should describe five (05) points on how our bodies defend themselves against
any harm which can be caused by micro-organisms in our health

INTRODUCTION (3 marks):
The student should give any relevant information on the body defence system
Body defence system is a system in the body which helps the body to prevent entrance of micro-organisms in the body. It involves some parts such as skin, eyes, ears, digestive system and blood.

MAIN BODY ((mention ½, explanation 1, example ½) 2marks@= 10)
The student should describe five 05 points on how the body defence system works
Skin: it acts as a barrier against entry of microorganisms. The glands on the skin like sweat

glands and sebaceous glands secrete sweat and sebum oily which contains antibacterial agents that kills bacteria

Eyes: the parts of the eyes like eyelashes, eyelids and eyebrows prevent the entry of dust which contains micro-organisms like bacteria. Tears glands also secrets tears which contains antibacterial agents

Ears: the outer ear canal produce wax material which traps bacteria, virus and small dust particles

Digestive system: Different regions of the digestive system produce and contain glands secretions which help in the body defence. Salivary glands for example secrets saliva which is a mild antibacterial agent, the harmless bacteria Which live in the mouth also protect it from various infectiy. In the stomach the hydrochloric acid secreted by stomach walls kill most of bacteria taken in with food. The intestinal juice contains antibacterial properties.

The blood: The blood is made up of white cells, lymphocytes and phagocytes which works together on defending the body. The lymphocytes provide antibodies that destroy pathogens while the phagocytes engulf and remove micro-organisms which invides the body.

Ciliated membrane, the presence of cilia in some parts of our bodies example in nose. The cilia trap dusts and bacteria which should be given out through sneezing and coughing.

CONCLUSION

Any relevant conclusion **(2 marks)**

10.Food preservation is the process of treating foodstuff with some materials to make food to last for long time.

(1 ½ mark)

(a)local traditional food preservation method

-drying, in this method moisture are removed from the food (dehydration method)

-Smoking, this method remove moisture from the foodstuff e.g fishes

- Curing, is the method where food is hung over charcoal fire and smoke-this kill most bacteria

(any 3 point (mention ½, explanation 1, example ½) @ 2mark= 6 marks)

(b) Advantages (local traditional)

-No chemicals are added to the food materials

-the food is preserved for a long time

-food is easily accessed in time of shortage

-it is not expensive because does not need much cash

(**any three point (mention ½, explanation 1, example ½) @ 2marks = 6 marks)**

Conclusion 1 ½ mark)

11. A student should describe the mechanism of hearing which enabled James to hear it's voice (**1mark)** A student should mention the organ responsible for hearing**(1 mark)**

i. Pinna collects sound waves from the surroundings and directs them to the middle ear through the external auditory canal or auditory meatus.

ii. The eardrum vibrates on receiving the sounds

iii. The vibrations of the eardrum cause disturbance to the ear ossiclesmalleous,incus and stapes which makes them vibrate and transmit the sound waves to the inner ear through the oval window.

iv. The vibrations of the oval window are transmitted to the cochlea and cause the fluid in the cochlea (perilymph) to vibrate and create up pressure and vibrations which

stimulate the sensory hair cells to create nerve impulse

v. The nerve impulse created will be sent to the brain by the auditory nerve for interpretation.

vi. The brain will interpret the impulse and send the feedback to which type of sound is that .

(6 points@2marks=12marks)

(Conclusion 1mark)